

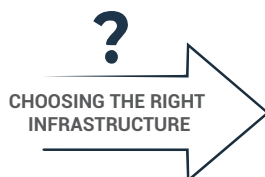
Self-Hosted vs Managed Cloud: Choosing the Right Infrastructure for Modern Apps

This comparison between a self-hosted and managed cloud will help you decide which one is ideal for your organisation.



SELF-HOSTED

CONTROL,
CUSTOMIZATION,
HIGH MAINTENANCE



MANAGED CLOUD

SCALABILITY,
RELIABILITY,
AUTOMATED UPDATES

A few years ago, building an online app meant buying physical servers, manually installing operating systems, and managing your own network and hardware maintenance. Today, you can deploy a globally distributed service in under 10 minutes. That's a wild shift.

But this has also created a genuinely hard decision for development teams: should applications run on self-hosted infrastructure, or should the team rely on managed cloud services? In a self-hosted infrastructure, the operational burden is on the organisation whereas in managed hosting, it is on the provider.

This isn't just a technical choice. It touches your cost model, your security strategies, long-term scalability, and how much control you want over your stack.

Self-hosting gives you full control and lower long-term costs, but your team is responsible for all the maintenance, security, and scaling. Managed cloud is faster to deploy and easier to operate but expect vendor lock-in and unpredictable bills at scale.

How we got here

To understand modern infrastructure, it is vital to understand how application deployment has evolved over the years.

Bare metal was the default: Early internet hosting meant dedicated physical machines. You installed Linux, configured networking by hand, and scaled by literally buying more servers. In short, you had complete control over hardware and software, but you needed to make a significant upfront investment. If you wanted to

scale an application, you needed to purchase additional servers, install them in data centres, and configure them manually. As the web application grew more complex, this model became difficult to maintain.

Then came virtualization:

Hypervisors like KVM and Xen changed the economics. Suddenly, one physical server could run multiple virtual machines. VPS providers like Linode and DigitalOcean were born from this. Customers could rent these servers, which reduced hardware costs and improved flexibility. You had access to a virtual server without buying or owning any hardware, which was a huge deal at the time.

Cloud platforms took it further:

The next major transformation came with large-scale cloud providers.